

Fetal Health Classification Using Machine Learning

Real-World Data Science Project - Healthcare Domain

Project Type	Multiclass Classification
Dataset	Fetal Health Dataset
Cleaned Observations	2,113
Predictor Variables	21
Target Variable	fetal_health
Final Model	Optimized Random Forest

1. Introduction

This project applies an end-to-end machine learning workflow to a healthcare dataset containing fetal cardiotocography-related measurements. The objective is to analyze the available variables and build a model capable of classifying the fetal health target into three classes.

The work includes data-quality assessment, exploratory analysis, preprocessing, model development, evaluation, cross-validation, hyperparameter optimization, and interpretation of model results.

2. Problem Statement

The goal of the project is to use the available fetal-health measurements to predict the multiclass target variable `fetal_health`. The project focuses on predictive classification within the provided dataset and does not claim to provide a medical diagnosis or replace professional clinical assessment.

3. Objectives

- Understand the structure and quality of the dataset.
- Identify and handle missing values, duplicate records, and other data-quality issues.
- Explore distributions, relationships, correlations, and class imbalance.
- Prepare training and testing data while avoiding data leakage.
- Train and compare multiple classification algorithms.
- Evaluate models using accuracy, precision, recall, F1-score, and confusion matrices.
- Use Macro F1-score as an important metric because the target classes are imbalanced.
- Validate and optimize the strongest model using stratified cross-validation.
- Identify influential predictors using Random Forest feature importance.
- Summarize findings, limitations, and possible future improvements.

4. Dataset Description

The original dataset contained 2,126 rows and 22 columns. Twenty-one columns are predictor variables and `fetal_health` is the target variable. Thirteen complete duplicate rows were identified and removed, resulting in a cleaned dataset containing 2,113 observations.

No missing values or infinite numerical values were found in the dataset.

Target Class Distribution

Class	Count	Percentage
1.0	1,646	77.90%
2.0	292	13.82%
3.0	175	8.28%

The distribution shows a clear class imbalance. Because Class 1.0 represents most observations, accuracy alone could provide a misleading impression of model quality. Macro-averaged metrics were therefore used alongside class-specific precision, recall, and F1-score.

5. Data Cleaning and Exploratory Analysis

Data cleaning included checking for missing values, duplicates, infinite values, target classes, and feature uniqueness. The 13 duplicate records were removed. Potential statistical outliers were examined but were not automatically deleted because unusual physiological measurements may represent meaningful observations rather than data-entry errors.

Exploratory analysis examined feature distributions, boxplots, correlations, relationships between features and the target, class-wise feature behavior, and correlations among predictor variables. The analysis showed differences in feature distributions and indicated that some variables had stronger associations with the encoded target than others.

6. Data Preprocessing

The predictor variables were separated into X and the fetal_health target into y. The cleaned dataset was divided into 80% training data and 20% testing data using a fixed random state of 42.

Stratified sampling was used to preserve approximately the same class proportions in the training and testing subsets. StandardScaler was fitted only on the training data and then applied to the testing data for scale-sensitive models. This approach prevents information from the test set from leaking into preprocessing.

Scaled features were used for Logistic Regression and K-Nearest Neighbors, while the original feature values were retained for Decision Tree and Random Forest models.

7. Model Development

Five classification approaches were evaluated: a most-frequent baseline classifier, Logistic Regression, K-Nearest Neighbors, Decision Tree, and Random Forest. The baseline established the minimum performance that meaningful machine learning models should exceed.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Random Forest	0.9527	0.9522	0.8785	0.9105
Decision Tree	0.9338	0.8982	0.8713	0.8843
Logistic Regression	0.8889	0.8214	0.7613	0.7885
K-Nearest Neighbors	0.8889	0.8447	0.7225	0.7720
Baseline Classifier	0.7801	0.2600	0.3333	0.2922

Random Forest achieved the strongest initial performance, with 95.27% accuracy and a Macro F1-score of 0.9105. The baseline classifier achieved 78.01% accuracy because of the dominant majority class, but its Macro F1-score was only 0.2922. This demonstrates why accuracy alone was not sufficient for this problem.

8. Cross-Validation and Hyperparameter Optimization

The Random Forest model was further evaluated using five-fold Stratified K-Fold cross-validation on the training data. Macro F1-score was used as the optimization objective so that all three classes contributed equally to model selection.

RandomizedSearchCV was used to explore combinations of Random Forest hyperparameters. The best configuration was:

- n_estimators: 500
- max_depth: 20
- min_samples_split: 2
- min_samples_leaf: 2
- max_features: sqrt
- class_weight: balanced

Best cross-validation Macro F1-score: 0.8881

9. Final Model Evaluation

The optimized Random Forest was evaluated on the held-out test set after model selection. It achieved 94.80% accuracy, Macro Precision of 0.9271, Macro Recall of 0.8944, and a Macro F1-score of 0.9092.

Class	Precision	Recall	F1-score	Support
1.0	0.96	0.98	0.97	330
2.0	0.88	0.76	0.81	58
3.0	0.94	0.94	0.94	35

Class 1.0 showed very strong performance. Class 3.0 also achieved high precision and recall despite its smaller test support. Class 2.0 was the most challenging category, with recall of 0.76 and an F1-score of 0.81.

The optimized model's test Macro F1-score of 0.9092 was slightly lower than the original Random Forest score of 0.9105. This small difference is acceptable because hyperparameters were selected through cross-validation on the training data rather than by repeatedly optimizing against the held-out test set.

10. Feature Importance

The optimized Random Forest was used to examine model-based feature importance. The ten most influential predictors in the fitted model were:

Rank	Feature	Importance
1	abnormal_short_term_variability	0.135134
2	percentage_of_time_with_abnormal_long_term_variability	0.123337
3	histogram_mean	0.102264
4	histogram_median	0.084076
5	accelerations	0.078813
6	mean_value_of_short_term_variability	0.069914
7	mean_value_of_long_term_variability	0.059876
8	prolongued_decelerations	0.053226

Rank	Feature	Importance
9	histogram_mode	0.052561
10	baseline value	0.044449

Abnormal short-term variability was the most influential predictor, followed by the percentage of time with abnormal long-term variability and histogram mean. These importance values describe how strongly the fitted Random Forest used each feature for prediction; they do not establish medical causation.

11. Key Findings

- The cleaned dataset contained 2,113 observations and 21 predictor variables.
- The target was imbalanced, with Class 1.0 representing 77.90% of the data.
- The low baseline Macro F1-score showed why accuracy alone was unsuitable for evaluation.
- Tree-based models outperformed Logistic Regression and KNN in this experiment.
- Random Forest was the strongest initial classifier with 95.27% accuracy and 0.9105 Macro F1.
- The optimized Random Forest achieved 94.80% accuracy and 0.9092 Macro F1 on the held-out test set.
- Class 2.0 was the most difficult class for the optimized model to identify.
- Variability-related and histogram-derived measurements were among the model's most influential predictors.

12. Limitations

- The target classes are imbalanced and the minority classes contain relatively few observations.
- Performance estimates may be affected by the limited dataset size.
- The analysis is restricted to the variables available in this dataset.
- Random Forest feature importance indicates predictive contribution, not causation.
- Independent external data would be required to assess generalization beyond this dataset.
- This project is a machine learning study and is not a substitute for professional medical assessment.

13. Future Work

- Evaluate the model on larger and independently collected datasets.
- Investigate methods to improve identification of Class 2.0 observations.
- Compare additional ensemble and boosting algorithms.
- Use repeated or nested cross-validation for more extensive model selection.
- Explore model-explanation methods for individual predictions.
- Study calibration and other application-relevant evaluation measures.

14. Conclusion

This project successfully developed an end-to-end machine learning workflow for multiclass fetal-health classification. Data cleaning, exploratory analysis, stratified splitting, appropriate feature scaling, multiple-model comparison, cross-validation, hyperparameter optimization, and feature-importance analysis were incorporated into the workflow.

Random Forest provided the strongest overall performance. The original model achieved 95.27% accuracy and a Macro F1-score of 0.9105. The cross-validation-optimized Random Forest achieved 94.80% accuracy and a Macro F1-score of 0.9092 on the held-out test set. Performance was strong for Classes 1.0 and 3.0, while Class 2.0 remained comparatively more difficult to identify.

Overall, the results demonstrate that the available measurements contain useful predictive patterns for classification within this dataset. However, these results should not be interpreted as clinical validation, and independent evaluation would be required before considering any real-world medical application.